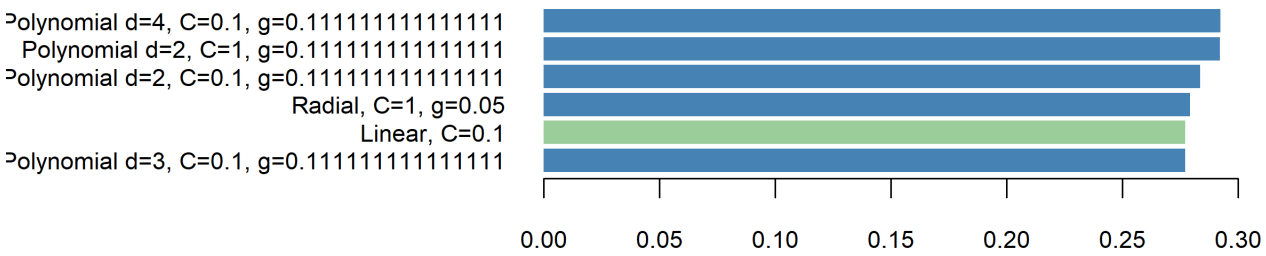
png
2

SVM Analysis: SAheart

Data and model selection. The response is binary `chd`; predictors are `sbp`, `tobacco`, `ldl`, `adiposity`, `famhist` (Absent = 1, Present = 2), `typea`, `obesity`, `alcohol`, and `age`. All 462 cases are complete (302 class 0, 160 class 1). I compared 18 SVM settings using common, stratified five-fold CV (seed 43209), standardizing each predictor within each training fold. All models used `C` in {0.1, 1, 10}; polynomial degrees 2–4 held `gamma = 1/9` and `coef0 = 1`; radial kernels used `gamma` in {0.05, 0.2}.

Choice and evaluation. The cubic polynomial (`C = 0.1`) had the smallest mean fold error (27.72%); the linear SVM (`C = 0.1`) was nearly identical (27.73%). Both had 128 pooled out-of-fold errors. Following the skill's one-error tie rule, I select the simpler linear SVM. The 5-by-5 nested estimate of the full tuning procedure was also 128/462 = 27.71%. Its out-of-fold confusion matrix was 257/45 for class 0 and 83/77 for class 1 (rows observed, columns predicted), giving 85.1% specificity but only 48.1% sensitivity. The error rate is below the 34.6% majority-class baseline, but many positive cases are missed; for high-stakes screening, a cost-sensitive criterion would be needed. The nested estimate comes from one seeded partition and has no reported confidence interval.

Candidate comparison



Kernel	C	gamma	Mean error	Errors	Selected
Polynomial d=3	0.1	0.111	0.2772	128	
Linear	0.1	-	0.2773	128	Yes
Radial	1.0	0.050	0.2793	129	
Polynomial d=2	0.1	0.111	0.2836	131	
Polynomial d=2	1.0	0.111	0.2923	135	
Polynomial d=4	0.1	0.111	0.2924	135	
Linear	1.0	-	0.2924	135	
Linear	10.0	-	0.2924	135	
Radial	1.0	0.200	0.2946	136	
Polynomial d=2	10.0	0.111	0.3032	140	
Radial	10.0	0.050	0.3033	140	
Polynomial d=3	1.0	0.111	0.3140	145	
Radial	10.0	0.200	0.3313	153	
Polynomial d=4	1.0	0.111	0.3334	154	
Polynomial d=3	10.0	0.111	0.3335	154	
Radial	0.1	0.050	0.3463	160	
Radial	0.1	0.200	0.3463	160	
Polynomial d=4	10.0	0.111	0.3487	161	

Appendix: Complete `SKILL.md`

```
---
name: svm-analysis
description: "Use only when explicitly invoked to assess classification data, compare and tune
SVM kernels, select a model, and report compact evidence."
disable-model-invocation: true
---

# SVM analysis

1. Confirm the response, predictors, class coding, sample size, missingness, class balance, and
predictor types. Flag data limitations before fitting.
2. Choose kernels to match plausible boundaries: linear for a roughly linear boundary; polynomial
for smooth interactions; radial for flexible nonlinear boundaries. Explain choices and keep the
candidate set small.
3. Tune cost `C` for every kernel. Tune kernel-specific parameters such as degree, radial
`gamma`, and polynomial offset when relevant; state all fixed settings and use a bounded,
justified search.
4. Use shared stratified folds for candidate comparison, or a justified alternative. Fit scaling,
imputation, feature selection, and other preprocessing on each training fold only.
5. Select by mean validation error using a metric appropriate to class balance and task costs.
Treat candidates within one pooled validation error of the minimum as practically tied; prefer
the simpler kernel, then lower cost. Refit the chosen pipeline on all training data. Use nested
CV or a separate test set for an unbiased final performance estimate; do not present tuned CV
error as independent test performance.
6. Include one compact figure comparing candidate validation scores and one compact table listing
settings, scores, and the selected model. Make the figure and table support, not replace, the
rationale.
7. Interpret the selected fit, report the key numerical results and uncertainty or limitations,
and distinguish selection scores from independent evaluation.
8. Prioritize only evidence needed to justify the final choice. Keep the analysis report to no
more than one printed page at 11-point font; verify the rendered page count. Append the complete
skill after that page when requested.
```